

[ARCHIVED CATALOGUE]

Quantum Information Science (MS)

The Master of Science in Quantum Information Science program requires a minimum of 28 graduate units with at least 20 units at the 500 level. A minimum grade point average of 3.0 must be earned on all course work applied toward the degree. This average must also be achieved on all 400-level and above course work attempted at USC beyond the bachelor's degree. A maximum of 4 units of Directed Research (590 or 790) may be counted toward the degree. A maximum of 4 transfer units count as credit (CR) toward the master's degree and are not computed in the grade point average.

Foundations (8 units)

- [EE 520 Introduction to Quantum Information Processing](#) Units: 4
- [PHYS 513 Applications of Quantum Computing](#) Units: 4

Core (11-12 units)

Take at least three courses from this list:

- [CSCI 575 Quantum Computing and Quantum Cryptography](#) Units: 4
- [EE 514 Quantum Error Correction](#) Units: 4
- [EE 515 Quantum Sensing: Machine Learning, Inference and Information](#) Units: 4
- [EE 589 Quantum Information Theory](#) Units: 4
- [PHYS 550 Theory of Open Quantum Systems](#) Units: 4
- [PHYS 559 Quantum Devices](#) Units: 4
- [PHYS 660 Quantum Information Science and Many-Body Physics](#) Units: 3

Electives (8-9 units)

- [CHEM 544 Introduction to Quantum Chemistry](#) Units: 4
- [CHEM 545 Theory and Practice of Molecular Electronic Structure](#) Units: 4
- [CHEM 555 Computational Quantum Chemistry: Methods and Applications](#) Units: 2, 3, 4
- [CSCI 513 Autonomous Cyber-Physical Systems](#) Units: 4
- [CSCI 556 Introduction to Cryptography](#) Units: 4
- [CSCI 570 Analysis of Algorithms](#) Units: 4
- [CSCI 596 Scientific Computing and Visualization](#) Units: 4
- [CSCI 653 High Performance Computing and Simulations](#) Units: 4
- [DSCI 510 Principles of Programming for Data Science](#) Units: 4
- [EE 503 Probability for Electrical and Computer Engineers](#) Units: 4
- [EE 506 Semiconductor Physics](#) Units: 4
- [EE 507 Micro- and Nano-Fabrication Technology](#) Units: 4
- [EE 508 Hardware Foundations of Machine Learning](#) Units: 4
- [EE 510 Linear Algebra for Engineering](#) Units: 4
- [EE 539 Engineering Quantum Mechanics](#) Units: 4
- [EE 540 Introduction to Quantum Electronics](#) Units: 4
- [EE 553 Computational Solution of Optimization Problems](#) Units: 3

- [EE 565 Information Theory and Its Application to \(Big\) Data Sciences](#) Units: 4
- [EE 577a VLSI System Design](#) Units: 4
- [EE 577b VLSI System Design](#) Units: 4
- [EE 589 Quantum Information Theory](#) Units: 4
- [EE 590 Directed Research](#) Units: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
- [EE 598 Electrical Engineering Research Seminar](#) Units: 1
- [PHYS 438a Introduction to Quantum Mechanics and its Applications](#) Units: 4
- [PHYS 438b Introduction to Quantum Mechanics and its Applications](#) Units: 4
- [PHYS 500 Graduate Colloquium](#) Units: 1
- [PHYS 516 Methods of Computational Physics](#) Units: 3
- [PHYS 558a Quantum Mechanics](#) Units: 3
- [PHYS 558b Quantum Mechanics](#) Units: 3
- [PHYS 590 Directed Research](#) Units: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
- [PHYS 660 Quantum Information Science and Many-Body Physics](#) Units: 3
- [PHYS 668 Advanced Quantum Mechanics](#) Units: 3
- [PHYS 678 Relativistic Quantum Field Theory](#) Units: 3
- [PHYS 680 Advanced Quantum Field Theory](#) Units: 3

Note:

Courses taken to satisfy the core requirement may not be double counted as electives.

Minimum Units Required: 28
